MFT Market Making Notes

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Market making is one machine with five P&L terms. At your latencies (about 750 ms to Hyperliquid, 2–5 ms to Binance) the workable setup is passive on Hyperliquid with a fast hedge on Binance, and at base-tier fees, fees matter more than speed.

Latency levers, cross-venue liquidity and the Prosperity analysis: Latency levers & Prosperity

How holding time and the exit change the economics: Holding time & exits

Two shapes of alpha search

Signal mining is a lottery; market making is tuning one machine. Both look like "find an edge in data", but the search works differently, and that changes where your hours should go.

Signal mining. You draw candidate signals (a factor, a predictor, a lead-lag relationship), test each, and keep the rare survivor. The draws are close to independent: a failed momentum variant tells you little about a funding-rate signal. Most draws are noise, so the more you test, the more false winners you find by luck. That is why this literature fixates on multiple testing. Harvey, Liu and Zhu (2016) argue a new equity factor should clear a t-statistic above 3.0, not 2.0. Bailey and López de Prado's Deflated Sharpe Ratio raises the bar for a strategy according to how many trials came before it.

Market making. Every variant (tight or wide, symmetric or skewed, passive on the DEX or on the CEX) earns or loses through the same five terms, set out in the next section. A change to one term is judged on the same fills, the same metric and mostly the same code. A better fair-value model is not a new strategy: it improves every strategy you could run. Progress compounds along one path instead of arriving as separate wins.

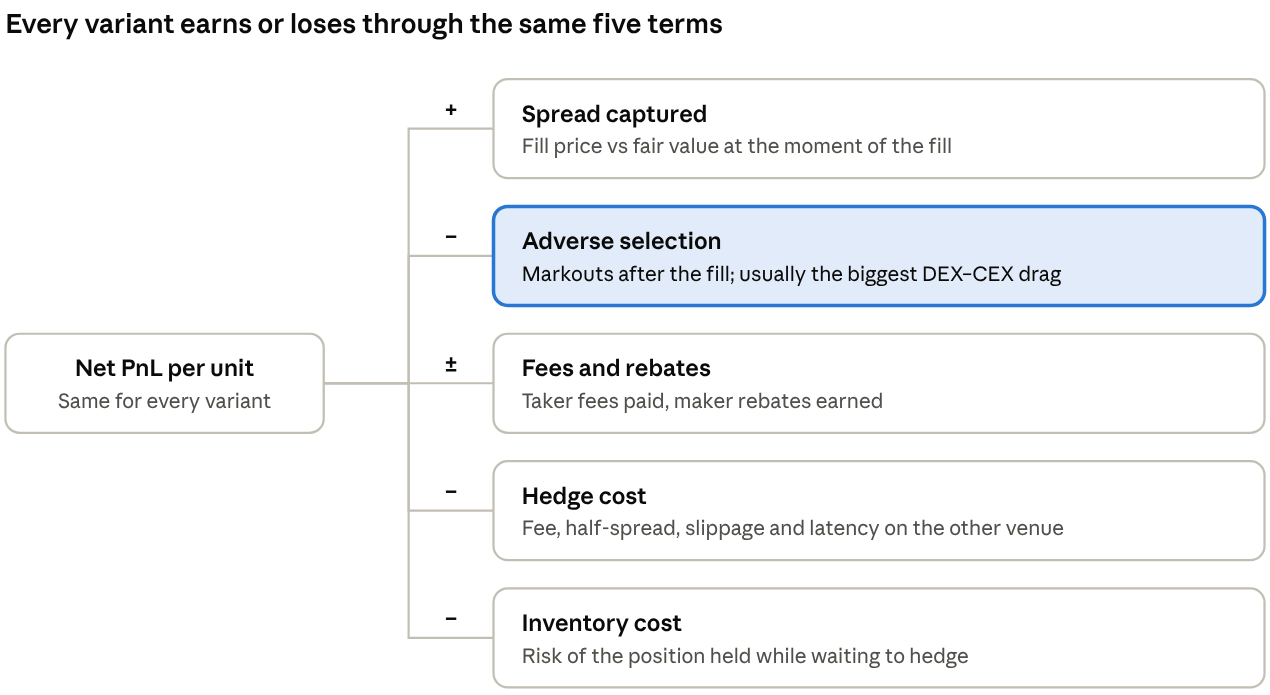
	Signal mining	Market making
Unit of search	An independent candidate signal	A change to one component of one strategy
What a failure teaches	Mostly that this signal is noise	Which term is costing you, and by how much

	Signal mining	Market making
Main statistical risk	False discoveries from many trials	Overfitting parameters on one replay; simulator error
What "throughput" means	Ideas tested per week	Time from a change to its measured effect
Question to ask weekly	How many ideas did I test, honestly counted?	Which term is the biggest drag, and did my last change move it?

The machine: five terms

Every market-making strategy, however it quotes, reduces to the same net P&L per unit traded:

```
\text{Net P\&L} = \text{spread captured} - \text{adverse selection} -
\text{fees} + \text{rebates} - \text{hedge cost} - \text{inventory cost}
```



market-making P&L · five terms

Every quoting idea is a change to one of these boxes; the highlighted one is where DEX–CEX books usually bleed.

- **Spread captured.** The distance between your fill price and your fair value at the moment of the fill. Measure it against fair value, not the mid, or a bad fair-value model hides inside this term.
- **Adverse selection.** How far price moves against you after you are filled. You get filled most when you are wrong, so fills are not a random sample. A live experiment on the Binance BTC perpetual by Albers, Cucuringu, Howison and Shestopaloff (2025) found maker fill likelihood negatively correlated with the next short-term return, with back-of-queue fills almost always losing. The standard measure is the markout curve, below.
- **Fees and rebates.** Edges here are measured in fractions of a basis point. In that same experiment, the average maker fill returned about −0.8 bp before a 0.5 bp rebate, leaving about −0.3 bp. A naive two-sided quoter lost heavily.
- **Hedge cost.** What it costs to lay off risk on the other venue: taker fee, half-spread, slippage and the price move during your latency.
- **Inventory cost.** The risk of holding a position while you wait. Avellaneda and Stoikov (2008) handle it with a reservation price that shifts against your inventory and a spread that widens with volatility.

Markout at horizon h, for a fill of side s (+1 buy, −1 sell) at price p, with fair value or mid P:

```
m_h = s \cdot \left( P_{t+h} - p \right)
```

Plot the average across fills for h from milliseconds to minutes, split by venue, side, and passive versus aggressive. A curve that falls after the fill is adverse selection; one that recovers is noise you were paid to absorb.

The Albers paper is itself a good example of the one-machine idea. The authors did not find a new strategy. They kept the same quoter and attacked one term, adverse selection, by quoting only when a classifier predicted a short-term reversal. That turned a losing book into a profitable one at minimum size.

Why the search compounds

In market making, the throughput that matters is iteration speed: how long from "I changed X" to "I can see what X did to each of the five terms".

- **Components are shared.** Fair value, the fill model, the hedge rule and the inventory policy sit under every variant. Improve one and every variant benefits.

- **Experiments are dependent.** The same fills inform all variants, and the same replay can score a new skew rule or a new hedge trigger against identical market conditions. One good dataset answers many questions.
- **Loop time sets your ceiling.** A two-week loop gives about 25 iterations a year; a one-day loop gives about 250. Shortening the loop usually beats adding ideas.

Two caveats keep this honest:

1. **Multiple testing does not disappear; it moves.** Sweeping parameters on one replay is many trials, and the Deflated Sharpe logic applies to parameter sweeps just as it does to strategy search. Log every variant you try.
2. **The machine has lottery-shaped parts inside it.** The Albers reversal classifier used 173 candidate features: that sub-problem is signal mining. The point is not that market making avoids signal search. It is that each signal earns its place only by moving a named term of one strategy, judged on one metric.

DEX–CEX: the same machine, one sharper term

In a DEX–CEX spread, adverse selection usually dominates, because one venue discovers price first and the other quotes stale prices for a moment.

- **LVR is the textbook case.** Milionis, Moallemi, Roughgarden and Zhang show that a hedged AMM liquidity provider earns trading fees minus "loss-versus-rebalancing" (LVR): a running cost from trading at worse-than-market prices, growing with volatility and with how much liquidity sits near the price. It is the five-term equation for passive DEX liquidity, with adverse selection given a closed form. For an order-book DEX the same logic holds by analogy: resting DEX quotes that lag the CEX get picked off.
- **Know which side of the arb you are on.** If the CEX leads, your passive DEX quotes are the stale price that arbitrageurs hit. Wu, Sui, Thiery and Pai (2025) estimate about $233.8M extracted by 19 CEX–DEX searchers on Ethereum from August 2023 to March 2025, with three searchers taking about three quarters. They tie searcher profit to integration with block builders. Your competitors on that leg may be concentrated and close to block production.
- **Simulation lies most on the DEX leg.** Queue position, block timing and inclusion, latency, and hedge slippage are hard to replay. The hftbacktest queue-model tutorial chooses its queue model by comparing backtest against live results. Albers et al. found randomly placed orders filled far more often than historical order data implied, because historical orders are placed by informed traders. Small live tests are how you calibrate the simulator, not a final step.

What changes in practice

Reframe each hypothesis as a change to one term, and judge it on the decomposition.

1. **Decompose a naive version first.** Before anything clever, run a simple quoter in replay and report each of the five terms. Work on the largest drag.
2. **Write hypotheses as component changes.** Not "a lead-lag signal is profitable", but "skewing DEX quotes by the CEX return over the last N ms cuts adverse selection at horizon h by X bp without cutting fill rate by more than Y%". The kill result is stated in that term.
3. **Report every experiment the same way.** Change in each term, markout curves by venue, side, and passive versus aggressive. One page, same layout every time, so results stack.
4. **Track loop time as a metric.** How long a change takes to reach a measured markout. Spend tooling effort on the step that dominates it, usually a replay harness that scores variants on identical data.
5. **Count trials anyway.** Keep a log of every parameter set tried, and deflate the winner.
6. **Go live small, early.** A tiny live book calibrates fill and queue assumptions. Without it, every term downstream of the fill model is a guess.

Binance–Hyperliquid: how fast do you need to be?

Not fast in the HFT sense. At base-tier fees, the fee term is several times larger than the cost of being slow.

What medium-frequency market making is. HFT market making wins by being first: requoting on every tick, holding inventory for fractions of a second. Medium frequency gives up that race on purpose:

- **Fair value** from Binance (usually the price leader) plus the Hyperliquid book, updated every 100 ms to 1 s.
- **Quotes** on Hyperliquid at fair value ± a half-spread wide enough to survive being stale, skewed by inventory.
- **Requotes** only when fair value moves past a threshold, not on every tick.
- **Inventory** held for seconds to minutes, hedged on Binance when it breaches a limit.
- **Edge** from wider spreads on coins the big makers don't crowd, and mean reversion in the Hyperliquid–Binance basis, not from queue position.

The stale window sets the speed requirement: the time from a Binance move until your Hyperliquid cancel takes effect.

$$\text{stale window} \approx \text{Binance feed} + \text{compute} + \text{cancel to Hyperliquid} + \text{Hyperliquid processing}$$

- Hyperliquid's validators are reported to sit in AWS Tokyo (Glassnode); Sydney to Tokyo is about 102 ms round trip (Hats Network).
- Hyperliquid's docs quote 0.2 s median and 0.9 s p99 end-to-end for a co-located client. Glassnode measured a median of 884 ms from AWS Tokyo in March 2026, almost all server-side.
- Within a block, cancels are processed before GTC and IOC orders, so a cancel that makes the same block as a taker wins. Professionals can also pay ALO priority to jump ahead of same-price quotes from the previous 400 ms.

Setup	Rough stale window (median)	What it supports
Sydney	~50 ms feed + ~50 ms to Hyperliquid + 200–900 ms processing	Research and paper trading; live only with very wide quotes
Tokyo cloud VM	a few ms + 200–900 ms processing	Medium-frequency MM on wider-spread coins
Pro HFT (Tokyo, own node, priority fees)	same processing, but ahead within the block	Top-of-book competition on majors; don't fight here

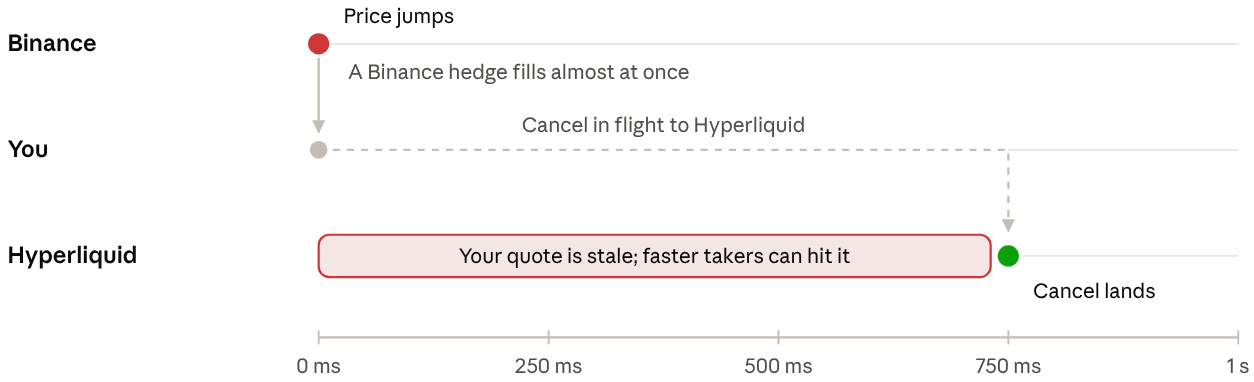
Why fees dominate. Assuming, for illustration, 80% annualised volatility, the price moves about 1.4 bp per $\sqrt{\text{second}}$: about 0.8 bp over 0.3 s and 1.4 bp over 1 s. Jumps matter more than diffusion, but the order of magnitude holds. At base tier, Hyperliquid maker is 0.015% and Binance futures taker is 0.05% (maker 0.02%). Quote on Hyperliquid and take on Binance costs about 6.5 bp per round trip; hedging passively cuts that to about 3.5 bp but adds inventory risk. That trade-off is the core design choice.

$$\text{half-spread captured} > \text{fees} + \mathbb{E}[\text{adverse move} \mid \text{fill}] + \text{hedge slippage}$$

Strategy at ~750 ms to Hyperliquid, 2–5 ms to Binance

Quote passively on Hyperliquid and hedge aggressively on Binance: your Binance speed is an asset for hedging, your Hyperliquid speed a liability for quoting.

Your Hyperliquid quote stays stale long after your Binance hedge is done



stale-quote window after a Binance move · to scale

The red band is the cost of being slow on Hyperliquid: everything else in this section is about pricing or shrinking it.

Four latencies, four jobs. On Hyperliquid the API returns the L1 execution result from a committed block, not just an acknowledgement, so ~750 ms is effectively time until your order or cancel is live.

Latency	Your number	What it controls
Hyperliquid order/cancel	~750 ms	How long your quote stays stale after Binance moves
Hyperliquid fill notification	unknown; measure it	How long before you know you need to hedge
Binance order	2–5 ms	How cleanly the hedge executes
Binance market data	probably low	How fresh your fair value is

Fill notification is the one people forget: learn of a Hyperliquid fill 500 ms late and the 2–5 ms hedge no longer saves you.

Relative speed is what matters. Glassnode's 884 ms public-API median suggests ~750 ms is roughly what an ordinary API user gets. Your exposure is to those faster than you (own nodes, priority fees). The cancel-first rule only helps when your cancel makes the taker's block; at ~750 ms you will usually be blocks behind.

Design rules at these speeds:

- **Price the stale window into the spread.** Cover the typical Binance move over ~0.75 s (about 1.2 bp diffusive at 80% vol), plus jumps, plus ~6.5 bp round-trip fees at base tier.
- **Requote rarely.** Every replace takes ~750 ms to land; move only when fair value crosses a threshold.
- **Widen before trouble, not after.** You cannot react to a jump in time, so pull back ahead of predictable volatility: Binance vol spikes, thinning books, scheduled events, funding times.
- **Layer depth.** Quotes a few ticks from fair value get picked off far less than top-of-book.
- **Hedge with an inventory band.** Take on Binance when the band is breached; hedge passively below it if you can carry the risk to save ~3 bp.
- **Never take on Hyperliquid to arb a mispricing.** Your IOC lands after faster arbitrageurs have taken it.

Expect P&L of many small spread captures punctuated by pick-offs when Binance jumps. Whether captures outweigh pick-offs is the whole question.

Measure first

Two numbers decide whether the Binance–Hyperliquid book can work, and both come from data you already have or can get cheaply.

1. **Hyperliquid fill-notification latency.** It fixes your real hedge delay. The docs and Glassnode differ by about 4× on order latency, so measure from your own setup rather than trusting either.
2. **Markout curve of Hyperliquid fills against the Binance mid, 0–2 s.** A sharp drop inside the first ~750 ms is the stale window costing you, and its depth tells you how wide to quote.

Also pin down: your firm's fee tier on both venues (it may matter more than anything above), and for two or three candidate coins, the Hyperliquid quoted spread against annualised volatility. Spread minus round-trip fees is the room left for adverse selection.

Questions to work through

These are left open on purpose; answer them in your own notes before reading further.

- [ ] For a naive quoter on your pair, which of the five terms do you expect to be the biggest drag, and what is the cheapest check that could prove you wrong?
- [ ] Which venue leads price discovery for your pair, and at what horizon? What does that imply about which venue you should be passive on?
- [ ] Is your DEX leg an AMM or an order book? Does LVR apply directly, or only as an analogy, and what changes if it is the latter?
- [ ] What is your current loop time from a change to a measured markout? Which single step dominates it?
- [ ] Looking back at your hypotheses so far, which were changes to one machine and which were independent lottery tickets?
- [ ] Where, if anywhere, does block-timing work enter the five-term equation?
- [ ] What is the smallest live sample that would let you check your fill model against reality?

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- Dwellir. How Hyperliquid's infrastructure works.
- Blockworks Research. Hyperliquid priority fees.
- Binance. Futures fee structure.
- Hats Network. Sydney to Tokyo latency.